

The Enduring Impact of Prenatal Exposure to Racial Riots on the Black Population*

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October 6, 2026

Abstract

This paper provides causal evidence on the long-term health consequences of prenatal exposure to racial riots in the United States. We construct a novel database linking historical riot events to individual-level mortality records and Census data, and estimate difference-in-differences models that exploit quasi-random variation in birth timing relative to riot events. Our design compares changes in longevity across adjacent nine-month cohorts exposed to anti-Black versus non-Black riots. We find that prenatal exposure to anti-Black riots reduces Black Americans' longevity by about 7.1 months. This effect is nearly twice the magnitude of the longevity reduction associated with prenatal exposure to lynchings documented in prior research, suggesting that the broader community disruption caused by riots—including housing loss, resource deprivation, and social dislocation—may have more severe long-run health consequences than individualized acts of racial terror. Our findings underscore the enduring biological imprint of structural racial violence and contribute to a growing literature documenting how early-life exposure to social shocks can shape long-term human capital and health outcomes.

Keywords: Racial Riots, Prenatal Exposure, Mortality, Longevity, Structural Violence, Early Life Shocks

JEL Classification: I14, I18, J15, J18, N31, N32, Z13

*We are grateful to Tiffany Green for her continued mentorship and unwavering support. We also thank Monica Garcia-Perez and seminar participants at ASHEcon 2025 for their helpful comments and discussions. This research was supported by the grant P30 AG017266, awarded to the Center for Demography of Health and Aging at the University of Wisconsin-Madison by the National Institute on Aging. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health. All errors are our own

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1 Introduction

Prior research demonstrates that in utero exposure to violent conflict, such as terrorism, civil unrest, or war, can have lasting effects on health, including increased risks of low birth weight, preterm birth, disability, and premature mortality ([Camacho, 2008](#); [Currie et al., 2022](#); [Lee, 2014](#); [Mansour and Rees, 2012](#); [Torche and Villarreal, 2014](#)). Black Americans are disproportionately exposed to structural and community-level violence, including excessive policing, residential segregation and displacement, and environmental injustice ([Boyd et al., 2020](#); [O’Flaherty and Sethi, 2019](#); [Vu et al., 2024](#)). Among the most extreme historical instances of such violence are racial riots, which involved large-scale destruction of Black neighborhoods, mass displacement, and lasting economic and social dislocation. Events like the 1919 Red Summer and the 1921 Tulsa Massacre were not isolated or spontaneous acts of disorder but organized and racially motivated assaults that devastated entire communities. While their historical significance is well documented, the long-term health consequences of these events remain largely unexplored in empirical research.

Recent work by [Vu et al. \(2023\)](#) offers compelling evidence that racial violence can have enduring biological effects. Their study found that prenatal exposure to lynchings, more localized and individualized acts of racial terror, reduced longevity among Black males by approximately 3.5 months. Building on this research, our study investigates whether prenatal exposure to racial riots, one of the most pervasive forms of racially motivated violence, has long-term consequences for individual health and longevity. While lynchings represented targeted acts of terror with symbolic significance for the broader community, riots directly affected far more individuals through physical violence, property destruction, and community displacement. These differences suggest that the mechanisms linking racial violence to long-term health outcomes might differ in important ways between lynchings and riots. Prenatal exposure to riots could affect fetal development not only through heightened maternal stress

but also through reduced access to essential resources such as housing and nutrition, as well as the erosion of community support systems. These compounded disruptions may produce more severe and lasting effects on health across the life course.

To examine these relationships empirically, we construct a novel database by linking historical riot events to individual-level mortality outcomes and Census data. In particular, our dataset combines information on 186 racial riots that occurred across the United States between 1900 and 1935 with mortality datasets from the CenSoc project and the 1940 Census. This allows us to identify individuals born in riot-affected communities during specific time windows and track their mortality outcomes later in life. We exploit variation in the timing of riots relative to individual birthdates to implement a difference-in-differences design. The first difference compares outcomes between individuals born 1–9 months after a riot event (exposed in utero to riots) and those born 10–18 months after a riot event. The second difference compares this temporal exposure gap across cohorts exposed to riots targeting Black communities versus those exposed to riots targeting other racial or ethnic groups.

Our main finding is that prenatal exposure to anti-Black riots is associated with a reduction in longevity of approximately 7.1 months among Black Americans. This effect persists across multiple specifications with varying sets of controls and fixed effects. Critically, this mortality impact is twice the magnitude of the 3.5-month reduction in longevity that [Vu et al. \(2023\)](#) found for in utero exposure to lynchings. This substantially larger effect aligns with our hypothesis that the more pervasive community disruption caused by riots would generate stronger health consequences than exposure to lynchings.

Our identification strategy relies on quasi-random variation in the timing of conception relative to riot events. We conduct extensive balance tests examining observable characteristics of parents across treatment and control groups, finding little evidence of systematic differences that might confound our estimates. We also conduct an omnibus test of whether the full set of parental characteristics jointly predicts riot exposure, conditioning on the

set of fixed effects. The results further support the idea that our empirical strategy, based on adjacent nine-month cohorts, plausibly produced treated and control groups that are uncorrelated with baseline characteristics after accounting for our set of fixed effects.

We next examine whether these effects reflect selective survival in utero and in infancy (culling) or lasting damage to development among survivors (scarring). Using county-level vital statistics, we find little evidence that anti-Black riots changed infant mortality, birth rates, or the racial composition of births, which suggests that selective survival did not substantially reshape exposed cohorts. In contrast, among individuals old enough to have completed their schooling by 1940, prenatal exposure is associated with roughly one fewer year of schooling and substantially lower socioeconomic and occupational status. These patterns are most consistent with scarring: prenatal exposure impaired development in ways that persisted into adulthood and ultimately shortened lives.

We provide several additional pieces of evidence that the estimated effect reflects prenatal exposure to riots rather than artifacts of the linked data or confounding shocks. First, in the spirit of an event study, we extend the sample to four additional nine-month placebo cohorts born 19–54 months after a riot, estimating each relative to the control cohort. These cohorts were not exposed in utero and serve as an analogue to the pre-period in a conventional event study. Estimates for all four placebo cohorts are small and statistically indistinguishable from zero. This pattern suggests that our results are not driven by differential attrition, mobility since birth, or truncation in the linked mortality and census records. Second, because our sample relies on linking Social Security Administration mortality records to the full-count 1940 census, we test whether riot exposure predicts successful linkage. The estimated relationship is negligible, roughly 0.2 percent of the mean linkage rate, which indicates that endogenous merging is unlikely to bias our estimates.

Our estimates are also robust to accounting for a range of contemporaneous shocks and investments that disproportionately affected Black Americans in the early twentieth-century

South. Because treatment is assigned by place of birth and the Great Migration relocated millions of Black southerners, with consequences for their longevity (Black et al., 2015), we include birth-state-by-1940-state-of-residence fixed effects to compare migrants and non-migrants separately. We further control for exposure to Rosenwald schools (Aaronson and Mazumder, 2011; Aaronson et al., 2021), for local Great Depression severity measured by the 1929–1933 decline in county bank deposits (Fishback et al., 2007), for exposure to malaria and hookworm eradication campaigns (Bleakley, 2007, 2010), and for the rollout of county health departments (Hoehn-Velasco, 2018). We allow each of these to affect Black cohorts differentially over time where relevant. Across all of these specifications, the estimated effect of prenatal riot exposure remains stable. Taken together, the evidence indicates that our findings are not an artifact of selective migration, differential access to schooling or medical care, or concurrent economic and epidemiological shocks.

This paper makes two contributions. First, we contribute to research in U.S. economic history on race and the Black experience. Prior studies document the long-run consequences of slavery and Jim Crow institutions (Sacerdote, 2005; Naidu, 2010; Althoff and Reichardt, 2024; Hornbeck and Logan, 2026), as well as of racial violence, racial politics, and backlash against Black advancement and migration (Jones et al., 2017; Derenoncourt, 2022; Chyn et al., 2024; Testa and Williams, 2026). Historical racial violence has been linked to reduced Black patenting (Cook, 2014), lower Black voter registration (Williams, 2022), higher contemporary unemployment and poverty (Williams et al., 2021), declines in Black homeownership and occupational status after the 1921 Tulsa massacre (Albright et al., 2021), and depressed property values and labor market outcomes after the riots of the 1960s (Collins and Margo, 2004, 2007). Our study contributes to this literature by providing causal evidence that in utero exposure to racial violence, specifically anti-Black riots, reduced longevity among affected Black cohorts. In doing so, we extend recent work by Vu et al. (2023), who document mortality effects of prenatal exposure to lynchings, by showing that riots, as a more

disruptive form of racial violence, had even more severe long-term health consequences.

Second, we construct a novel dataset on prenatal exposure to racial riots. Drawing on historical riot records ([Gilje, 1999](#); [Rucker and Upton, 2006](#); [Levy, 1991](#)), we identify racial riots in U.S. cities between 1900 and 1935 and link them by city and month of birth to individual death records from the CenSoc project. To our knowledge, this is the first dataset that follows individuals born in riot-affected communities from birth to old-age mortality.

The rest of the paper proceeds as follows. Section 2 provides historical background on racial riots in the early twentieth-century United States. Section 3 describes the riot data and the linked mortality and census records. Section 4 presents our identification strategy. Section 5 reports the main results, explores potential mechanisms, and presents placebo tests, checks for endogenous merging, and robustness exercises. Section 6 concludes.

2 Background

From 1900 through the 1930s, African American communities across the United States faced widespread White supremacist violence and “race riots” that caused mass death and destruction. This era, often called the “nadir” of American race relations, saw thousands of Black Americans lynched, burned, shot, and mutilated by white mobs ([National Archives, 2020](#)). The wave of violence culminated in episodic riots and massacres, such as the 1919 “Red Summer,” when White mobs attacked Black neighborhoods in more than two dozen cities. For example, Chicago’s 1919 race riot (the deadliest of that summer) left 38 people dead (23 African American, 15 white), injured over 500, and rendered roughly 1,000 Black families homeless ([Encyclopaedia Britannica, 2025](#)). Other events were even more brutal: in Elaine, Arkansas (September 1919) an estimated 100 Black men and women were killed ([National Archives, 2020](#)), and in the 1921 Tulsa Race Massacre Tulsa’s prosperous Greenwood district (“Black Wall Street”) was destroyed over 18 hours—some 35 city blocks burned

and up to 300 Black residents killed ([Oklahoma Historical Society, 2025](#)). These incidents occurred against the backdrop of Jim Crow segregation and a resurgent Ku Klux Klan, and they helped drive the Great Migration of Southern Blacks northward.

The immediate impacts of these riots were profound—homes, businesses, and lives were lost—and the long-run scars lingered for decades. Recent economic history research on specific events highlights lasting harm. For example, the 1921 Tulsa massacre devastated Black wealth: one study finds that Black household heads exposed to the riot were about 4.2 percentage points less likely to own homes in the ensuing decades, and that their average occupational status remained depressed even into the late 20th century ([Albright et al., 2021](#)). [Vu et al. \(2023\)](#) show that in utero exposure to the lynching of a Black victim led to a 3.5 month decrease in longevity among Black males. Taken together, these findings underscore how racial violence was not only a tool of terror in its time but also a structural force that shaped racial inequality in the United States.

3 Data

Mortality data.—The main data source for this study is the Social Security Administration’s Death Master File (DMF) and Numident data, which was accessed through the CenSoc Project website ([Goldstein et al., 2021](#)). The CenSoc project provides three data products. The DMF data reports deaths to male individuals over the years 1975–2005. The Numident data reports deaths to both males and females over the years 1988–2005. The Berkeley Unified Numident Mortality Database (BUNMD) reports deaths to both males and females with a high coverage for the years 1988–2007 and thin coverage for the years prior to 1988. The DMF and Numident data are linked to the 1940 census ([Ruggles et al., 2024](#)), which makes them ideal for this project. This linkage was conducted by us using the consistent individual-level unique identifier *histid*, which is available in both datasets and the 1940

census. This linkage enables us to infer individuals' county of birth, a critical variable for our analysis. We pooled the DMF and Numident data together to increase our power through an increase in sample size.

An advantage of our data is that they contain millions of observations before sample selection. Therefore, we can restrict the sample to narrowly defined cohorts (exposure or no exposure to racialized riots during gestation) while still retaining sufficient statistical power.¹ Another advantage of using the 1940-census-DMF-Numident linked sample is that it includes information on individuals' family characteristics and socioeconomic outcomes in 1940. This data allows us to explore possible mechanisms in later analyses.

As mentioned above, it is important to identify the county-of-birth to assign the local area a specific type of in utero exposure. The 1940 census contains information on county-of-residence in 1940 as well as the county-of-residence five years prior (i.e., in 1935) if the individual had moved. To increase the accuracy of the county-of-birth measure, we used the cross-census linking rules provided by the Census Linking Project ([Abramitzky et al., 2020](#)) to link data from a portion of individuals from the 1940 census back to their data from earlier censuses (1900–1930). We used the county information in the first census in which individuals were observed as the county-of-birth. For those who were not linked to any of the earlier censuses, we used county-of-residence in 1935 if applicable (if they had moved between 1935 and 1940) or the county-of-residence in 1940 as a proxy for county-of-birth. We are able to identify 76.2% of observations in historical censuses and use the 1940 county as a proxy for only 23.7% of observations. This method of assigning county of birth may introduce measurement errors that could potentially confound the estimates. However, two factors suggest it does not bias our estimates. First, as reported in [Section 5.3](#), we find little evidence that successful data linkage is correlated with riot exposure. Second, we account

¹While other data sets can be linked to the 1940 census, they have very small sample sizes. For example, the Health and Retirement Study provides an initial linked sample of slightly more than 9,000 individuals, which is too small to divide into narrower geographical, cohort, and specifically racial groups.

for potential cross-state mobility by including birth-state-by-state-of-residence (1940) fixed effects, which allows us to compare migrants and non-migrants separately. Results remain stable, indicating that cross-state mobility does not drive our findings (Figure 2).

Riot data.—Our main source of data on racial riots is the American Riots database. Historian Paul A. Gilje originally constructed this database for his book, *Rioting in America* (Gilje, 1999). Ryan Jones created a digital repository of these data, which Turchin (2012) used for analysis; we also use these digitized data for our study. Gilje defines a riot as “any group of twelve or more people attempting to assert their will immediately through the use of force outside the normal bounds of law.” Through an extensive review of news media and other sources, Gilje documents the circumstances of each riot, including the date, location (i.e., city), a description of the precipitating event, whether race played a role in the violence, and the outcome.² We cross-reference these data with other relevant sources on race riots, including Rucker and Upton (2006)’s data on race riots and the ICPSR political violence database (Levy, 1991). We focus on the period 1900–1935 because individuals born before 1900 are likely to be too old to appear in the 1975 death records.

We derived the main estimation sample using the following criteria. First, we excluded non-Black individuals. Second, we restricted the sample to four birth cohorts: Black births born in months 1–9 after each riot event against Black individuals, Black births born in months 10–18 after each riot event against Black individuals, Black births born in months 1–9 after each riot event against non-Black individuals, and Black births born in months 10–18 after each riot event against non-Black individuals.

Appendix Table A.1 reports summary statistics for the final sample. The average age at death in the final sample is 912.8 months (76 years). The average years of schooling in 1940

²In one such example, Gilje describes a 1917 event in Chester, PA where “July: Race riot. Began with the killing of a National Guardsman by a black man because the Guardsman insulted two black girls. Two days later rioting broke out that lasted three days. Three whites and three blacks killed. Many other injured. Many blacks left town.” Sources: *Crisis*, October 1917, p. 313; Downey and Hyser, *No Crooked Death*, p. 157; *NYT*, July 26, 1917.

among these individuals is 8.3 years.³ By design, approximately half of the individuals were exposed to a riot incident in utero, as each 9-month treatment cohort is paired with a 9-month control cohort. Also, by design, 100% of individuals were Black. In our final sample, 55.4% of individuals were in cohorts associated with riots targeting Black communities. Among all individuals, 25.3% were both exposed in utero to a riot and in a cohort associated with a riot targeting Black communities.

4 Empirical Strategy

Our identification strategy aggregates multiple difference-in-differences estimates to tease out the long-term impacts of in utero exposure to racialized civil unrest. For each riot event, we define narrowly spaced treatment and control cohorts of Black births within the same city. The treatment cohort comprises individuals born during the first to ninth month following a riot (months 1–9 post-event), while the control cohort includes those born in the subsequent nine-month window (months 10–18 post-event). By construction, these cohorts are locally defined in time and space, ensuring that they are likely to be subject to similar unobserved shocks and demographic trends, apart from the direct effects of the riot itself.

The primary threat to identification in this design is the systematic difference in birth timing across cohorts. Specifically, individuals in the treatment group are born nine months earlier than those in the control group, which may influence key outcomes such as residential mobility before the decennial census, match rates to the CenSoc mortality data, and truncation within the CenSoc mortality data. To address this concern, we construct an additional layer of comparison using births from the same states and same riot time periods but exposed to events not targeting Black communities. These “non-Black riot” events generate analogous treatment and control birth cohorts, allowing us to difference out any effects related to

³Individuals in some of these cohorts, specifically those born after 1920, had not completed their education in 1940.

the nine-month timing gap that are common across similarly timed cohorts exposed to civil unrest more broadly.

To implement this strategy, our empirical specification includes fixed effects for state-by-riot timing. The identifying variation thus comes from within-state comparisons between similarly timed Black cohorts differentially exposed to riots targeting Black individuals versus those exposed to riots targeting non-Black individuals. This approach relies on the assumption that Black births in these parallel cohorts are comparable along unobserved dimensions apart from the nature of riot targeting. In the subsequent section, we provide evidence on the plausibility of this assumption by examining observable parental characteristics.

If the effects of being born nine months earlier—including any associated seasonal variations in birth outcomes—are similar for Black births exposed to anti-Black riots and those exposed to riots not targeting Black communities, then the difference between these two groups effectively addresses concerns related to the treatment cohort being born nine months before the control cohort. Thus, our empirical strategy can be interpreted as a method of pooling evidence across 186 riot events to estimate the long-term impacts of in utero exposure to racially targeted unrest. Specifically, we estimate models of the following form:

$$\begin{aligned}
Y_{irct} = & \alpha_1 + \alpha_2 \text{Exposed In Utero}_t \times \text{Riot Targets Black}_r \\
& + \alpha_3 \text{Exposed In Utero}_t + \alpha_4 \text{Riot Targets Black}_r \\
& + \alpha_5 X_{irct} + \alpha_6 Z_{ct} + \xi_c + \zeta_t + \eta + \varepsilon_{irct}
\end{aligned} \tag{1}$$

Where Y_{irct} denotes the longevity of person i who was exposed to riot event r and born in city c during month-year t . $\text{Exposed In Utero}_t$ equals 1 for those born 1–9 months following a riot event in their city and 0 for those born 10–18 months after a riot event. $\text{Riot Targets Black}_r$ is an indicator variable that takes on a value of 1 if the riot is against Black communities and 0 if it is against another race/ethnicity.

The matrix X_{irct} includes parental education controls (0 years, equal or more than 5 years, equal or more than 9 years, some college, and missing) and paternal occupational score dummies (below the sample median, above the sample median, and missing). However, these controls are missing for a substantial portion of the sample. The city-level controls (represented by Z_{ct}) comprise several factors, including the proportion of the population in different age groups (11-18, 19-25, 26-55, >56 years old); the percentages of women, Black people, immigrants, and homeowners in the population; literacy rate, average occupational score, and proportion of families with children below age 5. In robustness checks, we control for state-by-cohort exposure to significant historical events affecting mortality, such as the Rosenwald school program and the Great Depression. We clustered the standard errors at the city level to account for serial correlation in error terms. The coefficient of interest is α_2 , which estimates the impact of exposure to a riot event during pregnancy among Black Americans.

City fixed effects (represented by ξ_c) control for both the observable and unobservable characteristics of each city that remain constant over time. Birth year-month fixed effects (represented by ζ_t) are included to capture time-invariant unobserved heterogeneity that might affect birth cohorts. As mentioned above, we include fixed effects for state-by-riot timing (represented by η), so identification relies on within-state comparisons of similarly timed Black cohorts who were differentially exposed to riots targeted against Black individuals versus those not.

In addition to mitigating concerns about differential matching rates to the mortality data and potential truncation due to age at death, our empirical specification is designed to account for both location-specific and time-specific shocks that are likely to affect Black communities similarly. By comparing adjacent cohorts within the same city, we also control for local shocks—such as economic or environmental changes—that may influence both cohorts similarly. In Section 5.3, we address additional potential sources of bias, including the

relatively low match rates between the DMF/Numident data and the 1940 Census.

It is important to note that our design exploits short-lived in utero exposure to local riot events to estimate causal effects on later-life outcomes. For each riot, we compare children born 1–9 months afterward (exposed in utero; “treated”) with those born 10–18 months afterward (not exposed; “controls”). Because exposure is transient, limited to a single birth cohort window, this setting differs from the staggered-rollout difference-in-differences frameworks emphasized in recent work (De Chaisemartin and D’Haultfoeulle, 2020; Goodman-Bacon, 2021; Sun and Abraham, 2021). Instead, it is an aggregation of multiple difference-in-differences estimates.

4.1 Identification Assumption

As discussed above, a key identifying assumption underlying our empirical strategy is that there are no systematic differences in the selection of individuals between the treatment and control groups. If there were differences in survival into adulthood by exposure to riots between, for example, children of different socioeconomic statuses, the models would provide biased estimates that reflect (in part) endogenous survival rather than solely riot exposure. To assess the plausibility of this assumption, we leverage data on observable family characteristics from the 1940 Census.

Table 1 reports estimates from regressions of parental characteristics on an indicator for in utero exposure to a riot, controlling for city-level covariates and fixed effects as specified in Equation 1. Across most outcomes, we find that the estimated coefficients are small in magnitude and statistically indistinguishable from zero. Two exceptions emerge: column 5 shows a higher rate of missing data on maternal education, and column 12 shows a lower proportion of fathers with below-median occupational income among Black children exposed in utero to riots targeting Black communities. The latter pattern suggests potential downward bias in our estimates, as higher parental socioeconomic status is generally associated

with improved health and longevity outcomes later in life (Almond et al., 2018; Currie, 2009; Hayward and Gorman, 2004; Montez and Hayward, 2011). However, these imbalances are not consistent across outcomes, weakening concerns about systematic selection.

To further support these balancing tests, we conduct an omnibus test of whether the full set of parental characteristics jointly predicts riot exposure, conditioning on the same set of fixed effects. The results, presented in Appendix Table A.2, reveal that nearly all coefficients are both economically negligible and statistically insignificant. The joint F-test confirms this lack of association (F-statistic = 1.398; p-value = 0.204). Overall, the results in this section support the idea that our empirical strategy, based on adjacent nine-month cohorts, plausibly produced treated and control groups that are uncorrelated with baseline characteristics after accounting for our set of fixed effects.

5 Results

5.1 Main Results

Panel A of Table 2 reports our estimates of Equation 1 for Black births. The model in column 1 includes city fixed effects, birth-year-month fixed effects, and birth-state-by-riot-time fixed effects. We then added parental controls and city-level controls in the models reported in columns 2 and 3, respectively. Across specifications, the estimated effect of prenatal exposure to riots remains stable, providing evidence of robustness to the inclusion of additional covariates. The fully parametrized model in column 3 indicates a 7.1-month shorter life span among Black individuals exposed to riots targeting Black communities during prenatal development.

To contextualize the magnitude of this effect, it is instructive to compare it with other well-documented early-life exposures. For example, Vu et al. (2023) examine the long-term effects of racialized violence in the form of lynchings and estimate that in utero exposure

to a lynching event reduces longevity among Black men by 3.5 months. Thus, the effect of riot exposure on longevity is approximately 90 percent larger than the estimated impact of lynching. This larger effect is consistent with the hypothesis that riots represent more widespread, community-level disruptions that may generate a broader set of stressors and material hardships during gestation, thereby compounding the adverse consequences for fetal development and long-term health.

Placebo test.—A natural concern is that the estimated effect reflects not the riot itself but some unobserved cohort- or city-specific changes in longevity that might coincidentally align with riot exposure. If so, the effect should also surface among residents who shared those cities and birth cohorts but were not the target of the violence. We therefore re-estimate Equation 1 on White births, who are excluded in the main analysis, and who we assume were unaffected by the riots targeting Black communities. Panel B of Table 2 returns coefficients that are small in magnitude and statistically insignificant. This strengthens our confidence that the observed negative effects on longevity are indeed specific to the Black population exposed to these riots, rather than being driven by broader, unobserved environmental factors.

5.2 Placebo Treatment Effects

A potential concern is that our findings are driven not by in utero exposure to the event itself, but by systematic differences in the linked mortality and census data, such as differential attrition, selective migration since birth, or truncation in the linked records across birth cohorts. To address this concern, we present graphical evidence for cohorts that were not exposed to the event in utero, following the spirit of an event study in Figure 1. Although these cohorts were born after the focal event rather than before it, they serve as an analogue to the “pre-period” in a conventional event study because they should not have been affected by prenatal exposure. We therefore expand the analysis beyond the original treatment and

control cohorts by including additional post-event birth cohorts. Specifically, we define the treatment cohort as individuals born 1–9 months after the event (i.e., those exposed in utero), and the control cohort as those born 10–18 months post-event. We then introduce four placebo cohorts: individuals born 19–27 months (placebo 1), 28–36 months (placebo 2), 37–45 months (placebo 3), and 46–54 months (placebo 4) following the event. Figure 1 plots the estimated coefficients and 95% confidence intervals for each cohort, with the control group normalized to zero. The estimates for all placebo cohorts are smaller in magnitude and statistically insignificant from zero. The insignificant effects on placebo cohorts mitigate concerns about differential attrition, mobility since birth, and truncation in the linked mortality and census data.

5.3 Endogenous Merging

The final sample is constructed through several steps of data merging, with the most critical step involving the linkage of Social Security Administration DMF/Numident data to the full-count 1940 census. A potential concern is the endogeneity introduced by this linking process. For instance, if people of lower socioeconomic status are less likely to be in the DMF/Numident-census linked sample and simultaneously have a higher probability of exposure to riots, the results may understate the true effects, as these people usually have, on average, lower longevity for other reasons.

To address this concern, we empirically test whether individuals with higher exposure to riots are less likely to appear in the DMF/Numident-census data. In so doing, we start with the original population from the full-count 1940 census and implement the same sample selection criteria as in Section 3. We then merge this dataset with our final sample at the individual level and create a dummy variable that indicates successful merging. We then regress this successful merging dummy on the exposure measures of Equation 1, conditional on covariates and fixed effects. The results, reported in Table 3, show an extremely small

point estimate, suggesting a change of 0.2% with respect to the mean (column 3). These results suggest that endogenous merging between the DMF/Numident data and the 1940 Census is unlikely to bias our main estimates in any meaningful way.

5.4 Robustness Checks

Figure 2 summarizes a series of robustness exercises, each designed to address a specific threat to the interpretation of our benchmark estimate. The first estimate reproduces our preferred model from column 3, Panel A of Table 2. The remaining specifications examine whether the estimated effect is robust to accounting for selective migration, contemporaneous investments in education and public health, or other time-varying economic and institutional changes that may have differentially affected Black communities during the same period.

Selective migration.—Treatment is assigned on the basis of city of birth, whereas mortality is observed decades later. During the period we study, the Great Migration relocated millions of Black southerners to northern cities, and the destination of migration was itself consequential for longevity: Black et al. (2015) find that Black men who left the South experienced an increase in mortality relative to otherwise similar men who remained. If riots induced differential out-migration among exposed cohorts, our estimates would conflate the effect of prenatal exposure with the effect of subsequent location. We therefore augment the specification with birth-state-by-state-of-residence-in-1940 fixed effects, so that we compare migrants and non-migrants separately. The results remain stable, suggesting that cross-state mobility does not bias our estimates.

Educational infrastructure.—Between 1914 and 1931, the Rosenwald Rural Schools Initiative financed the construction of nearly 5,000 schools for Black children across the South, producing substantial gains in attendance, literacy, cognitive test scores, and years of schooling (Aaronson and Mazumder, 2011). These gains plausibly translated into longevity: Aaronson et al. (2021) estimate that complete exposure to a Rosenwald school raised life expectancy

by two to three months. Because school construction was concentrated during the same decades in which riots occurred, differential exposure to Rosenwald schools across treatment and control cohorts would bias our estimates. We therefore add controls for exposure to Rosenwald schools. The estimated effects remain similar to the main results.

Macroeconomic shocks.—Our cohorts passed through infancy and early childhood during the Great Depression, whose severity varied sharply across localities and which is known to have moved infant mortality and fertility in U.S. cities (Fishback et al., 2007). If riot-affected cities were also more exposed to the banking collapse, and if Black residents bore a disproportionate share of that exposure, the resulting deprivation could be mistaken for an effect of prenatal riot exposure. Using county-level data from the Federal Deposit Insurance Corporation (1984), we calculate the decline in bank deposits between 1929 and 1933 and interact this measure with the Black population share and birth-year fixed effects, allowing Depression severity to affect Black cohorts differentially and flexibly over time. The coefficients are robust to adding these controls.

Public health campaigns.—The early twentieth-century South was also the site of two large disease-eradication efforts whose benefits accrued disproportionately to cohorts exposed in early life, and both were targeted at high-burden areas that overlap geographically with the states in which riots were concentrated. Malaria eradication in the 1920s generated lasting gains in income and cognitive attainment among exposed cohorts (Bleakley, 2010), and the Rockefeller Sanitary Commission’s hookworm eradication campaign of the 1910s raised school enrollment, literacy, later-life income, and later-life mortality in areas with high pre-campaign infection rates (Bleakley, 2007; Bleakley and Lange, 2009; Henderson, 2018; Noghanibehambari and Fletcher, 2026b). Failing to account for either would risk attributing to riots a cohort pattern generated by the timing of public health investment. We accordingly incorporate a state-specific malaria exposure index and the county-specific hookworm treatment rates, each interacted with the Black population share and birth-year

fixed effects. In both cases, our estimates remain robust.

County health departments.—Finally, we address another important change that reshaped the public health environment in the South during our sample period. The establishment of county health departments expanded access to affordable medical care and accounted for a meaningful share of the decline in rural mortality ([Hoehn-Velasco, 2018](#); [Noghanibehambari and Fletcher, 2026a](#)). Because the rollout of county health departments varied across locations and over time, it could generate cohort-specific differences in early-life conditions that coincide with the timing of racial riots. To address this concern, we construct a measure of exposure to county health departments and include it in our regressions. The inclusion of this control leaves our main results essentially unchanged.

Taken together, these exercises indicate that the estimated effect of prenatal riot exposure is not an artifact of selective migration, of differential access to schooling or medical care, or of contemporaneous economic and epidemiological shocks.

5.5 Mechanisms

The effects of prenatal exposure to anti-Black riots on longevity may operate through two broad mechanisms: culling and scarring. Culling occurs if the extreme stress and disruption associated with riots selectively increase fetal loss or infant mortality among the most vulnerable pregnancies, leaving a healthier surviving cohort. Scarring, in contrast, arises when prenatal exposure permanently alters fetal development among survivors, leading to worse health and socioeconomic outcomes throughout the life course. These mechanisms are not mutually exclusive, but they have different implications for interpreting our main estimates. If culling is substantial, then the surviving exposed cohort is positively selected, implying that our estimated effects on longevity understate the true consequences of prenatal exposure to anti-Black riots. If scarring dominates, then shorter adult lifespans reflect lasting damage sustained during fetal development rather than changes in cohort composition.

Historical accounts of anti-Black riots provide important context for understanding how racial violence can affect fetal development beyond direct physical harm. Unlike lynchings, which often targeted individual victims, anti-Black riots were collective episodes of racial terror that sought to intimidate, punish, and displace entire Black communities. Historical studies of major riots, including those in Atlanta (1906), East St. Louis (1917), Chicago (1919), and Tulsa (1921), document how white mobs attacked Black residents, destroyed homes and businesses, and disrupted community institutions, generating widespread fear and economic insecurity among Black families (Albright et al., 2021; Tuttle, 1996). These events were not confined to direct victims; violence spread through neighborhoods and was often widely reported in newspapers, exposing broader communities to psychological trauma and uncertainty (Albright et al., 2021; McWhirter, 2011).

For pregnant women, exposure to such pervasive threats and disruptions could generate severe emotional distress and physiological stress responses, including elevated cortisol, increased blood pressure, and heightened inflammation, that impair fetal development and increase risks of miscarriage, stillbirth, and other adverse pregnancy outcomes (Kinsella and Monk, 2009). Moreover, riots disrupted the housing, labor, food, and health-care conditions that support healthy pregnancies (Albright et al., 2021; Collins and Margo, 2007; Cook, 2014). Thus, anti-Black riots may operate through a culling mechanism by increasing fetal loss or infant mortality among the most vulnerable pregnancies, while also generating lasting developmental consequences among surviving children through scarring.

Both culling and scarring may arise through a common underlying pathway: maternal stress and the broader disruptions generated by racial violence. Sudden episodes of racial violence trigger physiological stress responses, including elevated cortisol, increased blood pressure, and heightened inflammation, that can impair fetal development and increase the risks of miscarriage, stillbirth, and infant mortality (Kinsella and Monk, 2009). A substantial literature connects prenatal exposure to violence and armed conflict with worse health at

birth and in later life (Camacho, 2008; Currie et al., 2022; Lee, 2014; Mansour and Rees, 2012; Torche and Villarreal, 2014). However, anti-Black riots likely affected pregnant women through channels extending well beyond the immediate psychological shock. Riots destroyed homes and businesses, displaced families, disrupted local labor markets, and reduced access to food, medical care, and other essential resources for weeks or even months afterward (Albright et al., 2021; Collins and Margo, 2007; Cook, 2014). A pregnant woman did not have to be directly targeted by the violence to feel it through lost earnings, displacement, and thinner nutrition and medical care. This broader reach is one reason to expect riots to leave a deeper imprint than the more targeted violence of a lynching, in line with the larger longevity effect we estimate.

Our data do not allow us to separate these channels directly. Maternal stress, the loss of housing, income, and nutrition, and the erosion of community support systems likely operated simultaneously, and historical records do not measure maternal health or household resources during pregnancy. We therefore focus on distinguishing culling from scarring, which have different implications for interpreting our estimates, and treat the relative importance of stress and material deprivation as a question our setting can inform only indirectly, through the comparison with lynchings.

To examine culling, we use county-level birth and infant mortality data compiled by Bailey et al. (2016). Because these historical vital statistics are available only at the county level, each riot is assigned to the county containing the affected city. We estimate event-study models for four demographic outcomes for birth cohorts spanning 1920 to 1935: the infant mortality rate, the birth rate, and the shares of births to White and Black mothers, absorbing birth-year, county, and riot-event fixed effects and weighting by county population. Figure 3 reports the resulting event-study estimates.

Overall, Figure 3 provides little evidence that anti-Black riots substantially altered the composition of exposed birth cohorts. Compared with 2–3 years prior to the incident, the

infant mortality rate and the birth rate are similar in the years following the riot as well as in prior years. For the share of births to Black birthing people, we do observe some increases after riots; however, all point estimates are statistically insignificant, limiting further interpretation.

To investigate scarring, we re-estimate Equation 1 using three adult outcomes observed in the 1940 Census: the socioeconomic index, years of schooling, and occupational prestige. Because educational attainment is only fully observed after early adulthood, the sample is restricted to individuals who were at least 25 years old in 1940. Table 4 reports the results. Prenatal exposure to an anti-Black riot is associated with a 6.64-point lower socioeconomic index, 0.97 fewer years of completed schooling, and a 4.09-point lower occupational prestige score. Relative to the sample means, these estimated differences correspond to approximately 41 percent, 12 percent, and 22 percent, respectively.

The estimates in Table 4 should be interpreted in light of the more selective sample used for these analyses. Compared with the longevity analysis in Table 2, the estimation sample is substantially smaller for two reasons. First, the outcomes examined here reflect completed adult attainment and are therefore only meaningful for individuals who had reached adulthood by the 1940 Census, requiring us to restrict the sample to individuals at least 25 years old in 1940 (born in 1915 or earlier). In contrast, age at death is obtained from subsequent mortality records and can be measured regardless of an individual's age in 1940. Second, the occupation-based measures, the socioeconomic index and occupational prestige, are observed only for individuals with occupations in the 1940 Census, excluding those who were unemployed or out of the labor force. Consequently, the occupation-based outcomes are available for fewer individuals than years of schooling. Given these sample restrictions, the estimates should be interpreted as evidence that exposure lowered adult attainment, not as exact measures of by how much.

Overall, Table 4 provides suggestive evidence consistent with the scarring mechanism.

Individuals exposed to anti-Black riots in utero attained substantially lower levels of education and socioeconomic status in adulthood, both of which are well-established predictors of later-life mortality (Chetty et al., 2016; Hayward and Gorman, 2004; Montez and Hayward, 2011). Together with the absence of meaningful changes in fertility, racial composition at birth, or birth rates documented in Figure 3, these findings suggest that the adverse effects of prenatal exposure primarily reflect persistent developmental scarring rather than selective survival. Although the historical data do not permit direct observation of maternal health or fetal conditions, the results are most consistent with prenatal scarring as the primary pathway, while recognizing that selective survival may also contribute to the estimated effects.

6 Conclusion

This paper provides new causal evidence on the long-term effects of racialized collective violence in the United States. By combining a dataset of racialized riots between 1900 and 1935 with individual-level mortality records from the CenSoc project and 1940 Census data, we estimate the effect of prenatal exposure to riots on longevity among Black Americans. Our difference-in-differences strategy exploits quasi-random variation in birth timing relative to riot events and compares changes in longevity across adjacent nine-month cohorts exposed to anti-Black versus non-Black riots. Our findings indicate that Black individuals exposed in utero to anti-Black riots experienced a reduction in longevity of approximately 7.1 months, nearly double the longevity loss recently estimated for prenatal exposure to lynchings (Vu et al., 2023). This pattern is consistent with the view that riots—through their broader geographic reach, greater material destruction, and large-scale disruption of community networks—produced more severe and enduring consequences than more targeted acts of racial terror.

More broadly, our findings highlight that the legacy of racialized violence is not limited to immediate economic or political disruption but may extend into the biological lives of the next generation. Understanding these long-term impacts is critical for assessing the full cost of racial violence and its contribution to persistent racial disparities in health and longevity.

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7 Tables

Table 1: Balancing Test: In Utero Exposure to Racial Riot and Observable Parental Characteristics

	Outcomes:										
	Mother's education zero	Mother's education 1-12 years	Mother's education college	Mother's education missing	Father's education zero	Father's education 1-12 years	Father's education college	Father's education missing	Father's occ. income score below median (9)	Father's occ. income score above median (10)	Father's occ. income score missing (11)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
[Exposed in Utero] $\times$ [Riot Targets Black]	0.0054 (0.0061)	-0.0261 (0.0178)	-0.0121 (0.0078)	0.0327** (0.0163)	0.0086 (0.0058)	-0.0078 (0.0153)	-0.0184* (0.0099)	0.0176 (0.0199)	0.0168 (0.0183)	-0.0429** (0.0196)	0.0261 (0.0184)
Observations	12,352	12,352	12,352	12,352	12,352	12,352	12,352	12,352	12,352	12,352	12,352
R-squared	0.0627	0.2779	0.0694	0.3011	0.0782	0.2068	0.0800	0.2276	0.1521	0.1130	0.2294
Mean of dependent variable	0.015	0.531	0.026	0.428	0.021	0.366	0.025	0.588	0.246	0.171	0.583
City fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Birth-year-month fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Birth-state-by-riot-time fixed effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
City-level controls	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Notes: This table examines whether there are observable differences in parental characteristics between the treatment and control groups. Specifically, we estimate parental characteristics as a function of in utero exposure to riots, while controlling for city fixed effects, year-month of birth fixed effects, and birth-state-by-riot-time fixed effects, following the specification outlined in Equation 1. Standard errors, clustered on city, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: In Utero Exposure to Racial Riot and Later-Life Longevity

	<i>Outcome: Age at Death (Months)</i>		
	(1)	(2)	(3)
<i>Panel A. Black births</i>			
[Exposed in Utero] $\times$ [Riot Targets Black]	-7.4588** (3.0798)	-7.2500** (2.9393)	-7.0575** (3.2499)
Observations	12,352	12,352	12,352
R-squared	0.4125	0.4138	0.4150
<i>Panel B. Placebo test using White births</i>			
[Exposed in Utero] $\times$ [Riot Targets Black]	-0.0704 (0.7788)	-0.0745 (0.7810)	-0.2968 (0.8626)
Observations	303,120	303,120	303,120
R-squared	0.2249	0.2254	0.2254
City fixed effects	✓	✓	✓
Birth-year-month fixed effects	✓	✓	✓
Birth-state-by-riot-time fixed effects	✓	✓	✓
Parental controls		✓	✓
City-level controls			✓

Notes: Panel A reports estimates from Equation 1 among Black births. Panel B reports estimates of Equation 1 among White births as a placebo test. Standard errors, clustered on city, are in parentheses. Parental controls include dummies for maternal education, paternal education, and paternal socioeconomic status. City covariates include share of population in different age groups, share of females, average number of children less than 5, share of Blacks, share of immigrants, share of homeowners, and literacy rate. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: Exposure to Riot and Endogenous Data Linking

	<i>Outcome: Successful Merging between 1940 Census and DMF Data</i>		
	(1)	(2)	(3)
[Exposed in Utero] $\times$ [Riot Targets Black]	-0.00439 (0.00443)	-0.00416 (0.00420)	0.00161 (0.00305)
Observations	19,065	19,065	19,065
R-squared	0.99632	0.99634	0.99976
Mean of dependent variable	0.652	0.652	0.652
Birth-state-by-riot-time fixed effects	✓	✓	✓
Parental controls		✓	✓
City-level controls			✓

Notes: This analysis tests whether individuals with higher exposure to riots are less likely to appear in the DMF-1940 census linked data. The dependent variable is a dummy indicating successful merging between the 1940 census and DMF data. Standard errors, clustered on city, are in parentheses. Regressions include birth-state-by-riot-time fixed effects and controls. Parental controls include dummies for maternal education, paternal education, and paternal socioeconomic status. City covariates include share of population in different age groups, share of females, average number of children less than 5, share of Blacks, share of immigrants, share of homeowners, and literacy rate. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

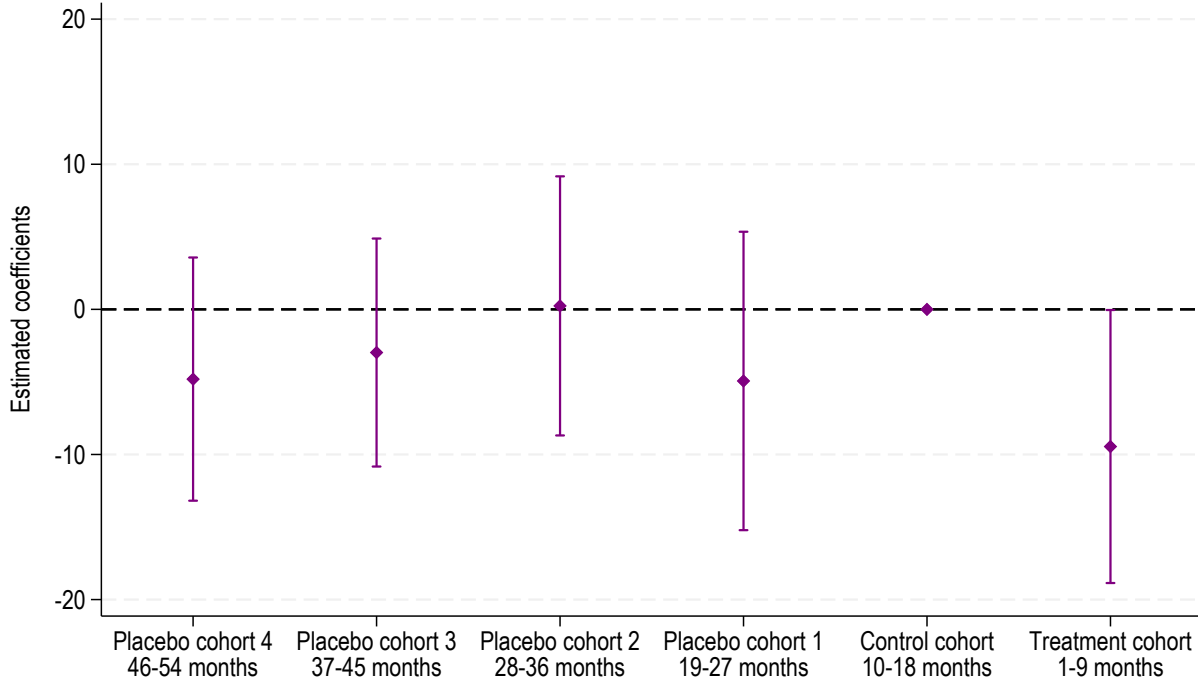
Table 4: Mechanism Channels

	Outcomes:		
	Socioeconomic score	Years of schooling	Occupational prestige score, Siegel
	(1)	(2)	(3)
[Exposed in Utero] $\times$ [Riot Targets Black]	-6.64** (3.20)	-0.97** (0.37)	-4.09** (1.76)
Observations	2,336	2,989	3,032
R-squared	0.16	0.24	0.10
Mean of dependent variable	16.13	7.84	19.01
City fixed effects	✓	✓	✓
Birth-year-month fixed effects	✓	✓	✓
Birth-state-by-riot-time fixed effects	✓	✓	✓

Notes: The dependent variables are individual-level measures from the 1940 Census: socioeconomic score, years of schooling, and the Siegel occupational prestige score. The Siegel occupational prestige score captures the perceived social standing or prestige associated with an occupation rather than its earnings or educational requirements. All regressions use the specification in Equation 1 but exclude parental and city-level controls. Standard errors, clustered on city, are in parentheses. The sample is restricted to individuals who were at least 25 years old in 1940 to ensure that educational attainment is fully observed. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

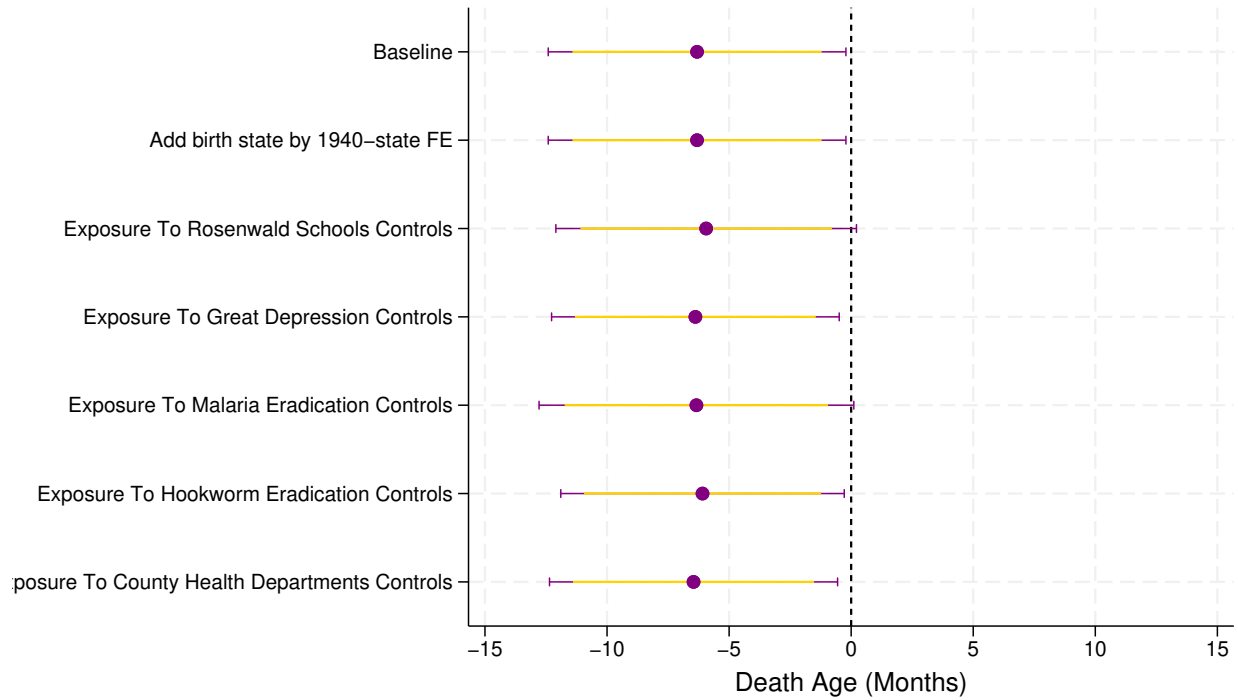
8 Figures

Figure 1: Event Study Style Results



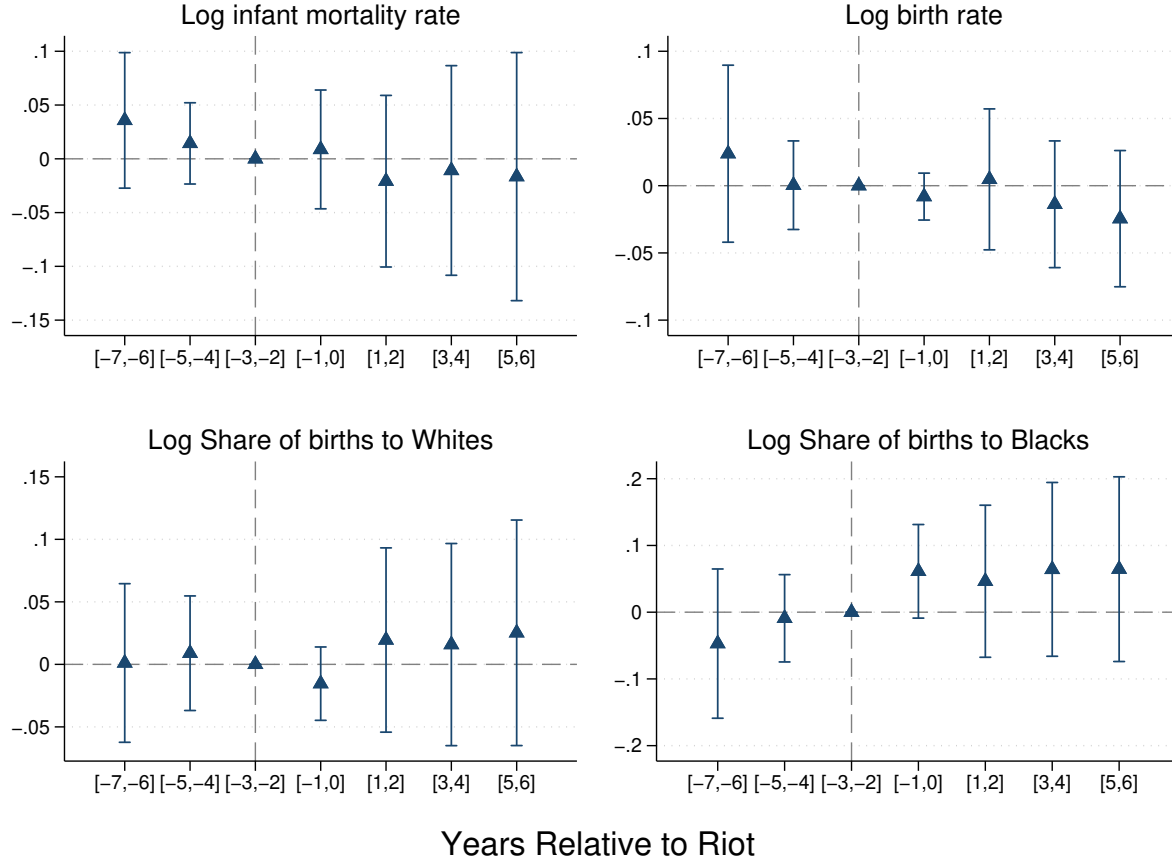
Notes: This figure presents graphical evidence of treatment effects for cohorts not exposed to the riot event in utero, in the spirit of an event study. Specifically, we expand the analysis beyond the original treatment and control cohorts by including additional post-event birth cohorts. We define the treatment cohort as individuals born 1–9 months after the event (i.e., those exposed in utero), and the control cohort as those born 10–18 months post-event. We then introduce four placebo cohorts: those born 19–27 months (placebo 1), 28–36 months (placebo 2), 37–45 months (placebo 3), and 46–54 months (placebo 4) following the event. Point estimates and 95% confidence intervals for each cohort are illustrated, with the control group normalized to zero. Standard errors are clustered on city. The outcome is age at death (in months). Regressions include city fixed effects, birth-year-month fixed effects, birth-state-by-riot-time fixed effects, and controls. Parental controls include dummies for maternal education, paternal education, and paternal socioeconomic status. City covariates include share of population in different age groups, share of females, average number of children less than 5, share of Blacks, share of immigrants, share of homeowners, and literacy rate.

Figure 2: Robustness Checks across Alternative Specifications



Notes: This figure presents robustness checks across alternative specifications. Point estimates and confidence intervals are illustrated. The whisker lines indicate 95% and the purple lines indicate 90% confidence intervals. Standard errors are clustered at the city level. The outcome is age at death (in months). Regressions include city fixed effects, birth-year-month fixed effects, birth-state-by-riort-time fixed effects, and controls. Parental controls include dummies for maternal education, paternal education, and paternal socioeconomic status. City covariates include share of population in different age groups, share of females, average number of children less than 5, share of Blacks, share of immigrants, share of homeowners, and literacy rate.

Figure 3: Examining the Effects on Fertility and Infant Mortality



Notes: This figure presents event-study estimates of the effect of racial riots on county-level fertility and infant mortality, constructed from county-level vital statistics on births and infant deaths (Bailey et al., 2016). Because the natality data are reported for counties rather than cities, each riot is assigned to the county containing the affected city. The panel covers birth years 1915 through 1935. The outcome variables are the log infant mortality rate, the log birth rate, and the log shares of births to White and to Black mothers. Event time is defined as the birth year minus the year of the riot and is grouped into two-year bins, with the bin covering three to two years before the riot normalized to zero. Regressions include birth-year, county, and riot-event fixed effects. Observations are weighted by county population. The figure reports point estimates with 95% confidence intervals, with standard errors clustered at the county level.

A Appendix

Table A.1: Summary Statistics

	Mean	SD	Min	Max
Age at death (months)	912.807	102.935	493	1228
Log age at death (years)	4.318	0.119	3.714	4.625
Age at death > 70 Years	0.735	0.441	0	1
Year of birth	1918.837	6.723	1900	1938
Year of death	1994.928	7.08	1975	2005
[Exposed in Utero] × [Riot Targets Black]	0.253	0.435	0	1
[Exposed in Utero]	0.471	0.499	0	1
[Riot Targets Black]	0.554	0.497	0	1
Non-Hispanic Black	1	0	1	1
Years of schooling	8.337	3.777	0	20
Socioeconomic index	17.335	16.098	3	92
Occupational income score	18.034	7.784	3	80
Mother's education zero	0.015	0.122	0	1
Mother's education 1-12 years	0.531	0.499	0	1
Mother's education college	0.026	0.158	0	1
Mother's education missing	0.428	0.495	0	1
Father's education zero	0.021	0.143	0	1
Father's education 1-12 years	0.366	0.482	0	1
Father's education college	0.025	0.156	0	1
Father's education missing	0.588	0.492	0	1
Father's occupational score missing	0.246	0.431	0	1
Father's occupational score below median	0.171	0.377	0	1
Father's occupational score above median	0.583	0.493	0	1
Observations	12,352			

Table A.2: Joint Balancing Test

	<i>Outcome: Exposed in Utero × Riot Targets Black</i>
	(1)
Mother's education zero	-0.00642 (0.01454)
Mother's education 1-12 years	-0.00338 (0.00658)
Mother's education college	-0.00896 (0.01233)
Father's education zero	0.08162** (0.03951)
Father's education 1-12 years	0.05492 (0.03562)
Father's education college	0.03841 (0.03880)
Father's occupational income score below median	-0.05681* (0.03244)
Father's occupational income score above median	-0.06130* (0.03530)
Observations	12,352
R-squared	0.76
<i>p</i> -value of joint significance test	0.204
F-statistic of joint significance test	1.398
City fixed effects	✓
Birth-year-month fixed effects	✓
Birth-state-by-riot-time fixed effects	✓

Notes: This table examines the correlation between exposure to a riot incidence and parental observable characteristics in the spirit of an omnibus test, conditional on a set of fixed effects. Standard errors, clustered on city, are in parentheses. Regressions include city fixed effects, birth-year-month fixed effects, and birth-state-by-riot-time fixed effects. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.